

Lattice QCD Data Summary Report

Chiral Condensates and Susceptibilities Across Finite Temperature

1 Definitions and Observables

1.1 Chiral Condensates

The bare chiral condensate on a lattice volume $V = N_s^3 \times N_t$ is defined as

$$\langle \bar{\psi}\psi \rangle_q = \frac{1}{V} \langle \text{Tr} (D_q^{-1}) \rangle, \quad q \in \{l, s\} \quad (1)$$

evaluated via stochastic Z_2 volume source vectors.

To eliminate the additive ultraviolet divergence proportional to m_q/a^2 as well as residual chiral symmetry breaking effects, the renormalized subtracted chiral condensate is defined as

$$\Delta_{l,s}^R(T) = \frac{1}{Z_m(\beta)} \left[\langle \bar{\psi}\psi \rangle_l - \frac{m_l + m_{\text{res}}}{m_s + m_{\text{res}}} \langle \bar{\psi}\psi \rangle_s \right] \quad (2)$$

where $m_{\text{res}}(\beta)$ is the residual quark mass for Domain Wall Fermions, and $Z_m(\beta)$ is the quark mass renormalization factor (with scalar density renormalization $Z_S = 1/Z_m$) providing multiplicative renormalization. In physical units of $[\text{GeV}^3]$, the physical chiral condensate is given by $\langle \bar{\psi}\psi \rangle_{\text{sub}}^R(T) = a^{-3} \Delta_{l,s}^R(T) = (N_t T)^3 \Delta_{l,s}^R(T)$.

1.2 Chiral Susceptibility

The light-quark disconnected chiral susceptibility χ_{disc} is defined on a four-dimensional volume $V = N_s^3 \times N_t$ as

$$a^2 \chi_{\text{disc}} = V \left[\left\langle \left(\frac{1}{V} \text{Tr}(D_l^{-1}) \right)^2 \right\rangle - \left\langle \frac{1}{V} \text{Tr}(D_l^{-1}) \right\rangle^2 \right] \quad (3)$$

The unbiased quadratic cross-product estimator for k stochastic noise vectors $O_{l,i} = \text{Tr}(D_l^{-1})_i$:

$$\bar{O} = \frac{1}{k} \sum_{i=1}^k O_{l,i}, \quad \overline{O^2} = \frac{1}{k(k-1)} \left[\left(\sum_{i=1}^k O_{l,i} \right)^2 - \sum_{i=1}^k O_{l,i}^2 \right] \quad (4)$$

In physical units of $[\text{GeV}^2]$, the disconnected chiral susceptibility is given by

$$\chi_{\text{disc}}(T) = (N_t T)^2 a^2 \chi_{\text{disc}} \quad [\text{GeV}^2] \quad (5)$$

where T is expressed in GeV.

2 Chiral Condensate Data and Temperature Dependence

2.1 Numerical Data Tables

Table 1: Renormalized subtracted chiral condensate in lattice units $\Delta_{l,s}^R$ and physical units $\langle\bar{\psi}\psi\rangle_{\text{sub}}^R$ [GeV³] across 8 temperature points on standard $48^3 \times 16$ lattices.

β	T [MeV]	Size ($N_s^3 \times N_t$)	N_{cfg}	$\Delta_{l,s}^R$	$\langle\bar{\psi}\psi\rangle_{\text{sub}}^R$ [GeV ³]
4.13	138.23	$48^3 \times 16$	2000	$(1.0608 \pm 0.0067) \times 10^{-3}$	$(1.1477 \pm 0.0073) \times 10^{-2}$
4.15	145.75	$48^3 \times 16$	2024	$(7.7109 \pm 0.0675) \times 10^{-4}$	$(9.7789 \pm 0.0856) \times 10^{-3}$
4.17	153.31	$48^3 \times 16$	2008	$(5.0728 \pm 0.0662) \times 10^{-4}$	$(7.4872 \pm 0.0978) \times 10^{-3}$
4.18	157.03	$48^3 \times 16$	2029	$(4.1649 \pm 0.0659) \times 10^{-4}$	$(6.6056 \pm 0.1045) \times 10^{-3}$
4.20	164.55	$48^3 \times 16$	2072	$(2.3896 \pm 0.0561) \times 10^{-4}$	$(4.3610 \pm 0.1023) \times 10^{-3}$
4.23	176.43	$48^3 \times 16$	2012	$(9.2886 \pm 0.3810) \times 10^{-5}$	$(2.0894 \pm 0.0857) \times 10^{-3}$
4.30	202.52	$48^3 \times 16$	2052	$(8.2043 \pm 1.3300) \times 10^{-6}$	$(2.7913 \pm 0.4512) \times 10^{-4}$
4.405	241.60	$48^3 \times 16$	4855	$(4.0303 \pm 0.2068) \times 10^{-7}$	$(2.3280 \pm 0.1195) \times 10^{-5}$

Table 2: Renormalized subtracted chiral condensate $\Delta_{l,s}^R$ and physical condensate $\langle\bar{\psi}\psi\rangle_{\text{sub}}^R$ [GeV³] for finite temporal extent ensembles at $\beta = 4.17$ ($m_l = 0.0020, m_s = 0.0400$).

Ensemble	Size	T [MeV]	N_{cfg}	$\Delta_{l,s}^R$	$\langle\bar{\psi}\psi\rangle_{\text{sub}}^R$ [GeV ³]
L48T18	$48^3 \times 18$	136.28	150	$(1.0555 \pm 0.0140) \times 10^{-3}$	$(1.5544 \pm 0.0206) \times 10^{-2}$
L40T16	$40^3 \times 16$	153.31	62	$(6.4908 \pm 0.3757) \times 10^{-4}$	$(9.5800 \pm 0.5546) \times 10^{-3}$
L32T14	$32^3 \times 14$	175.21	281	$(2.0385 \pm 0.2439) \times 10^{-4}$	$(3.0086 \pm 0.3600) \times 10^{-3}$
L32T12	$32^3 \times 12$	204.41	108	$(-2.1007 \pm 2.4182) \times 10^{-5}$	$(-3.1005 \pm 3.5691) \times 10^{-4}$

2.2 Plots vs Temperature

2.2.1 Standard $48^3 \times 16$ Ensembles

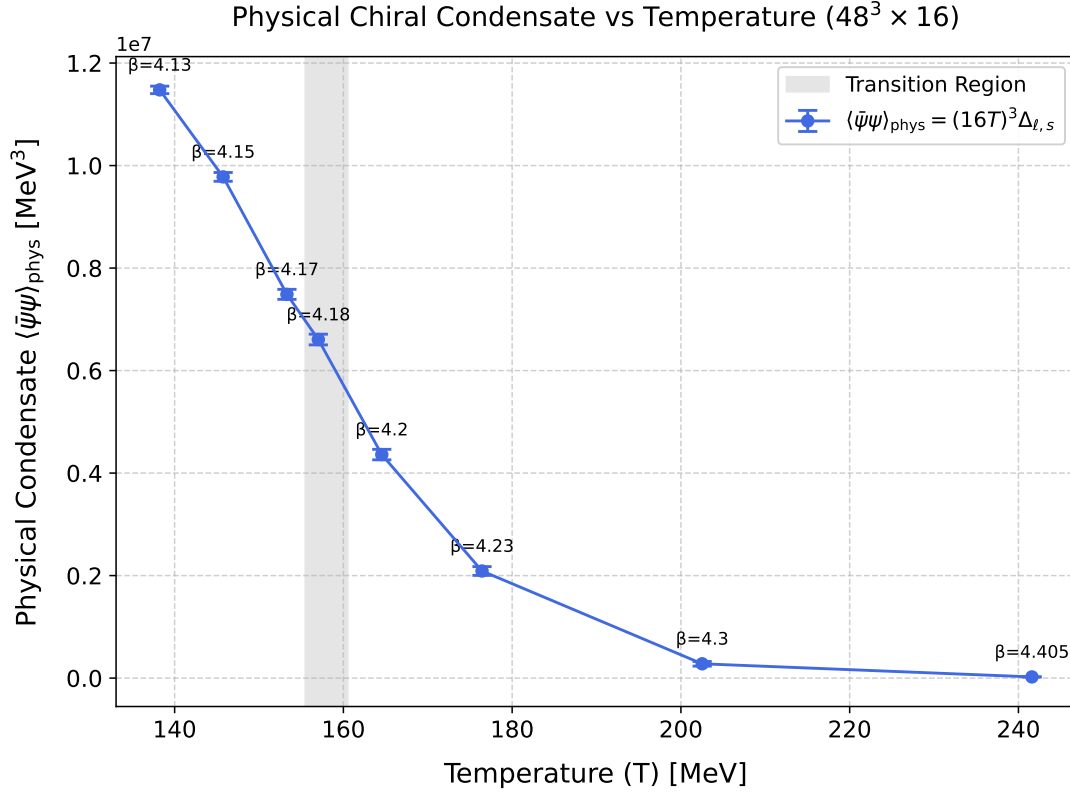


Figure 1: Physical renormalized subtracted chiral condensate $\langle \bar{\psi}\psi \rangle_{\text{phys}} = (N_t T)^3 \Delta_{l,s}^R [\text{MeV}^3]$ as a function of temperature T [MeV] for standard $48^3 \times 16$ lattices. The shaded band marks the crossover transition window $T \in [155, 158]$ MeV.

2.2.2 Finite Temporal Extent Ensembles ($\beta = 4.17$)

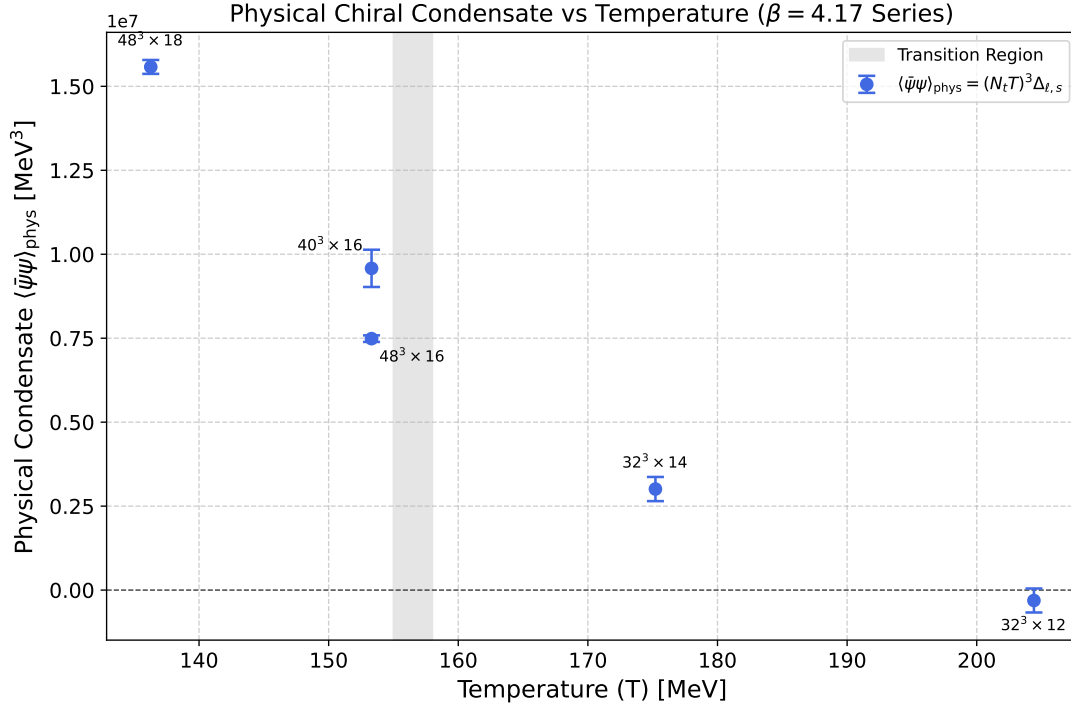


Figure 2: Physical renormalized subtracted chiral condensate $\langle \bar{\psi}\psi \rangle_{\text{phys}} = (N_t T)^3 \Delta_{l,s}^R$ [MeV³] across temporal lattice sizes $N_t = 12, 14, 16, 18$ ($T = 153.31 \times 16/N_t$ [MeV]) at fixed $\beta = 4.17$. The shaded band marks the crossover transition window $T \in [155, 158]$ MeV.

3 Chiral Susceptibility Data and Temperature Dependence

3.1 Numerical Data Tables

Table 3: Disconnected chiral susceptibility χ_{disc} in physical units of [GeV²] on standard $48^3 \times 16$ lattices (unbiased quadratic estimator without RM subtraction).

β	T [MeV]	Size ($N_s^3 \times N_t$)	N_{cfg}	χ_{disc} [GeV²]
4.13	138.23	$48^3 \times 16$	2000	0.6622 ± 0.0231
4.15	145.75	$48^3 \times 16$	2024	0.7800 ± 0.0267
4.17	153.31	$48^3 \times 16$	2008	0.8583 ± 0.0258
4.18	157.03	$48^3 \times 16$	2029	0.9150 ± 0.0323
4.20	164.55	$48^3 \times 16$	2072	0.7660 ± 0.0277
4.23	176.43	$48^3 \times 16$	2012	0.4091 ± 0.0232
4.30	202.52	$48^3 \times 16$	2052	0.0719 ± 0.0150
4.405	241.60	$48^3 \times 16$	4856	$(8.65 \pm 1.69) \times 10^{-5}$

Table 4: Disconnected chiral susceptibility χ_{disc} in physical units of [GeV²] for finite temporal extent ensembles at $\beta = 4.17$ ($m_l = 0.0020, m_s = 0.0400$).

Ensemble	Size	T [MeV]	N_{cfg}	χ_{disc} [GeV²]
L48T18	$48^3 \times 18$	136.28	151	0.3173 ± 0.0366
L40T16	$40^3 \times 16$	153.31	62	0.4863 ± 0.0803
L32T14	$32^3 \times 14$	175.21	281	0.4263 ± 0.0608
L32T12	$32^3 \times 12$	204.41	108	0.1361 ± 0.0523

3.2 Plots vs Temperature

3.2.1 Standard $48^3 \times 16$ Lattices

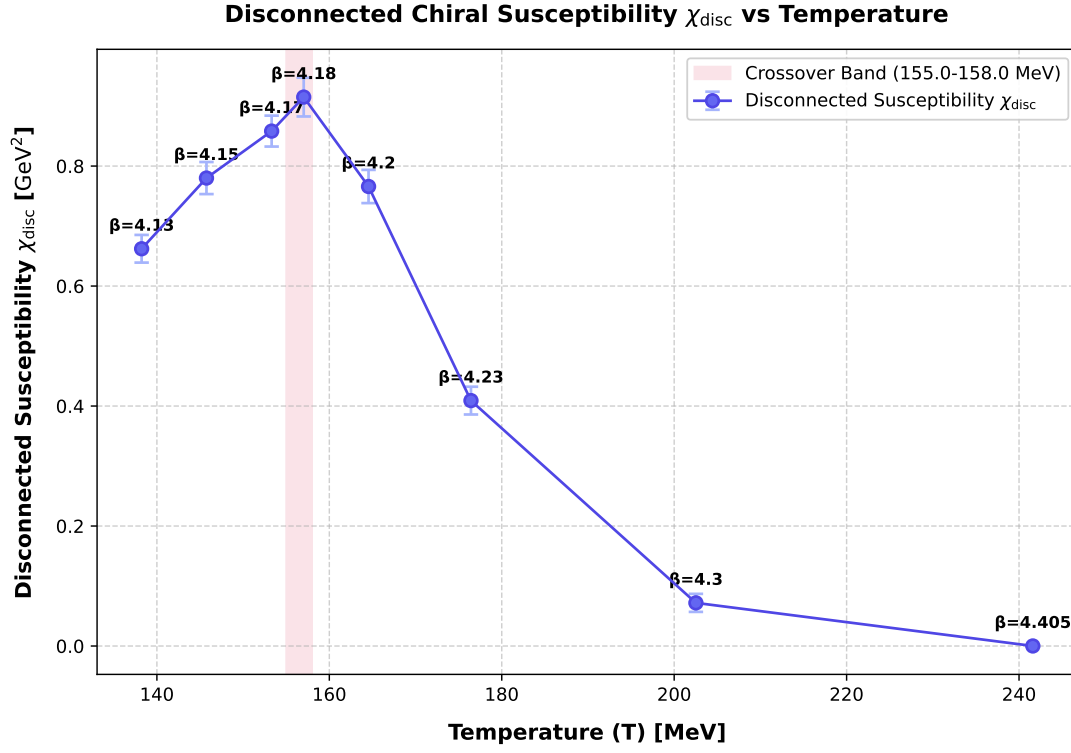


Figure 3: Disconnected chiral susceptibility χ_{disc} [GeV²] as a function of temperature T [MeV] for standard $48^3 \times 16$ lattices. The shaded band denotes the crossover window $T \in [155, 158]$ MeV.

3.2.2 Finite Temporal Extent Ensembles ($\beta = 4.17$)

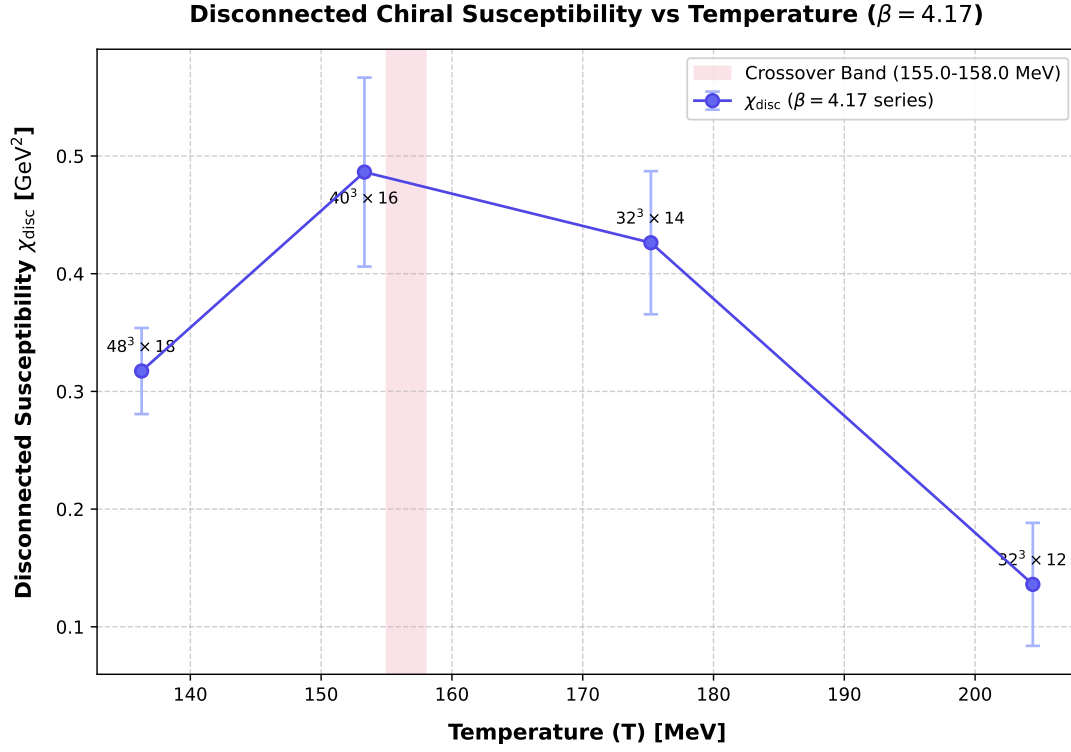


Figure 4: Disconnected chiral susceptibility χ_{disc} [GeV^2] as a function of temperature T [MeV] for finite temporal extent ensembles at $\beta = 4.17$ ($m_l = 0.0020$, $m_s = 0.0400$).

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